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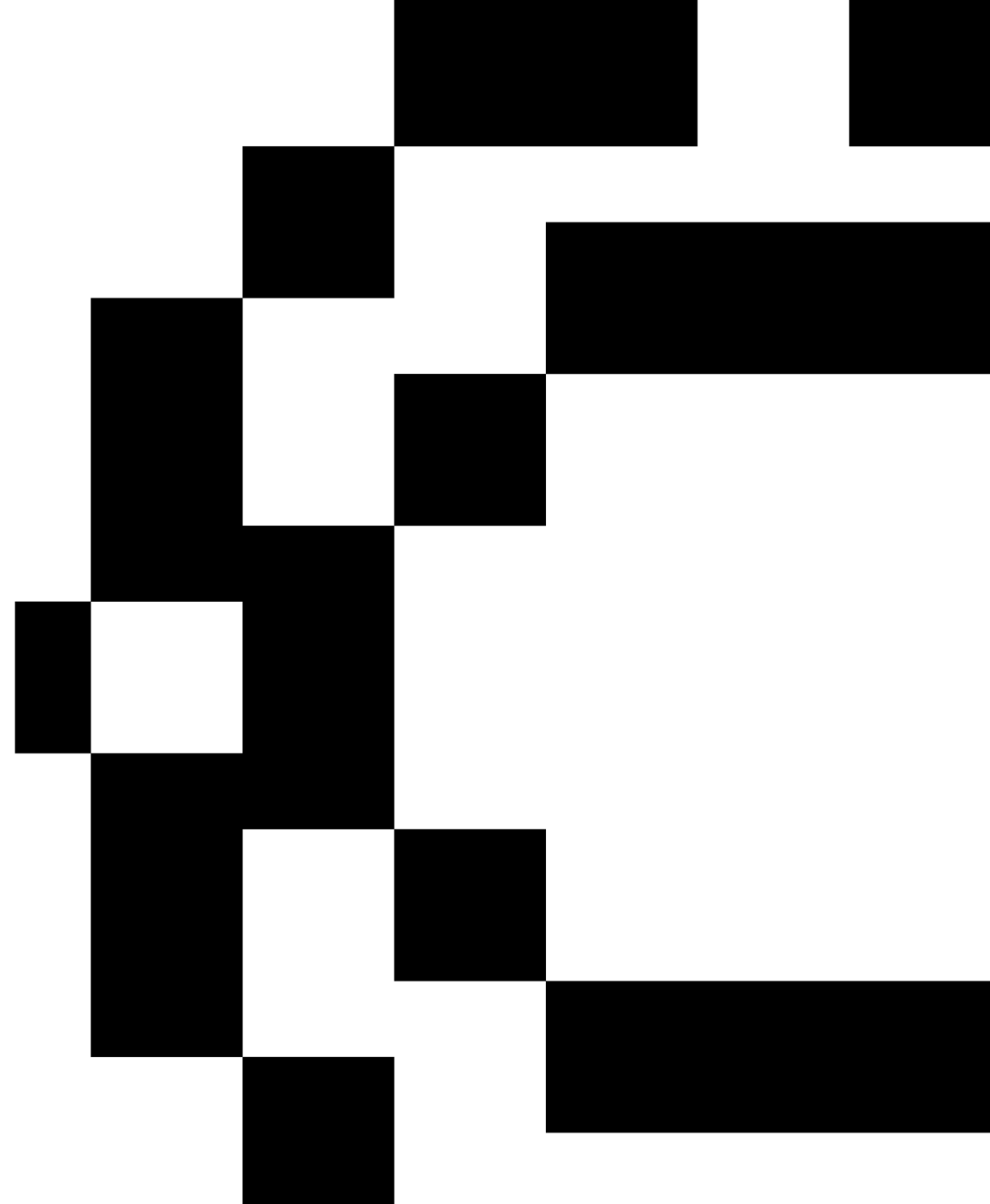
ZeroR, OneR, Naive Bayes and KNN

Data Mining II| MGI 2526

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Agenda

01

ZeroR



02

OneR



03

**Bayesian
Classifiers**



04

**Instance Based
Classifiers**

1. Algorithms

Simplicity First

Simple algorithms often work very well!

There are many kinds of simple structure, eg.:

- One attribute does all the work
- All attributes contribute equally & independently
- A weighted linear combination might do
- Instance-based: use a few prototypes
- Use simple logical rules
- ...

Success of method depends on the domain

01. What is ZeroR?

Classification simplest method **based only on the target ignoring all predictors**

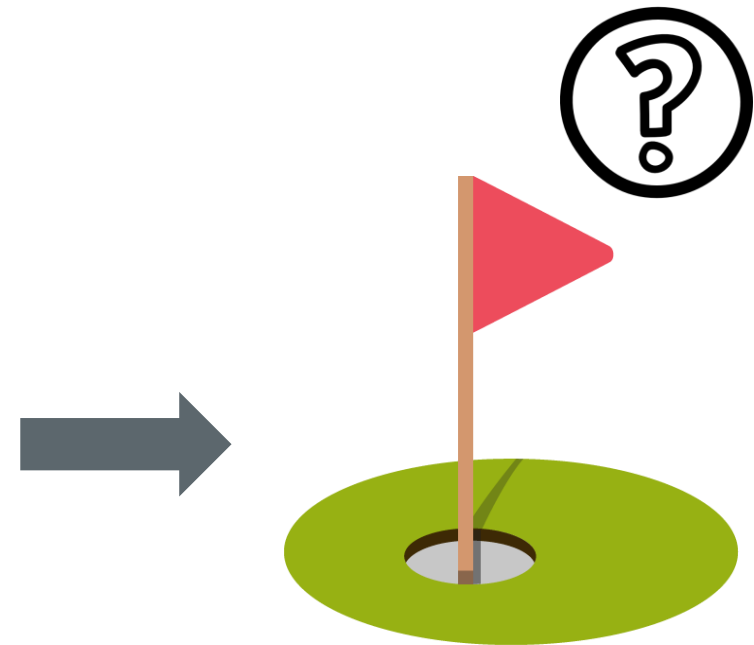
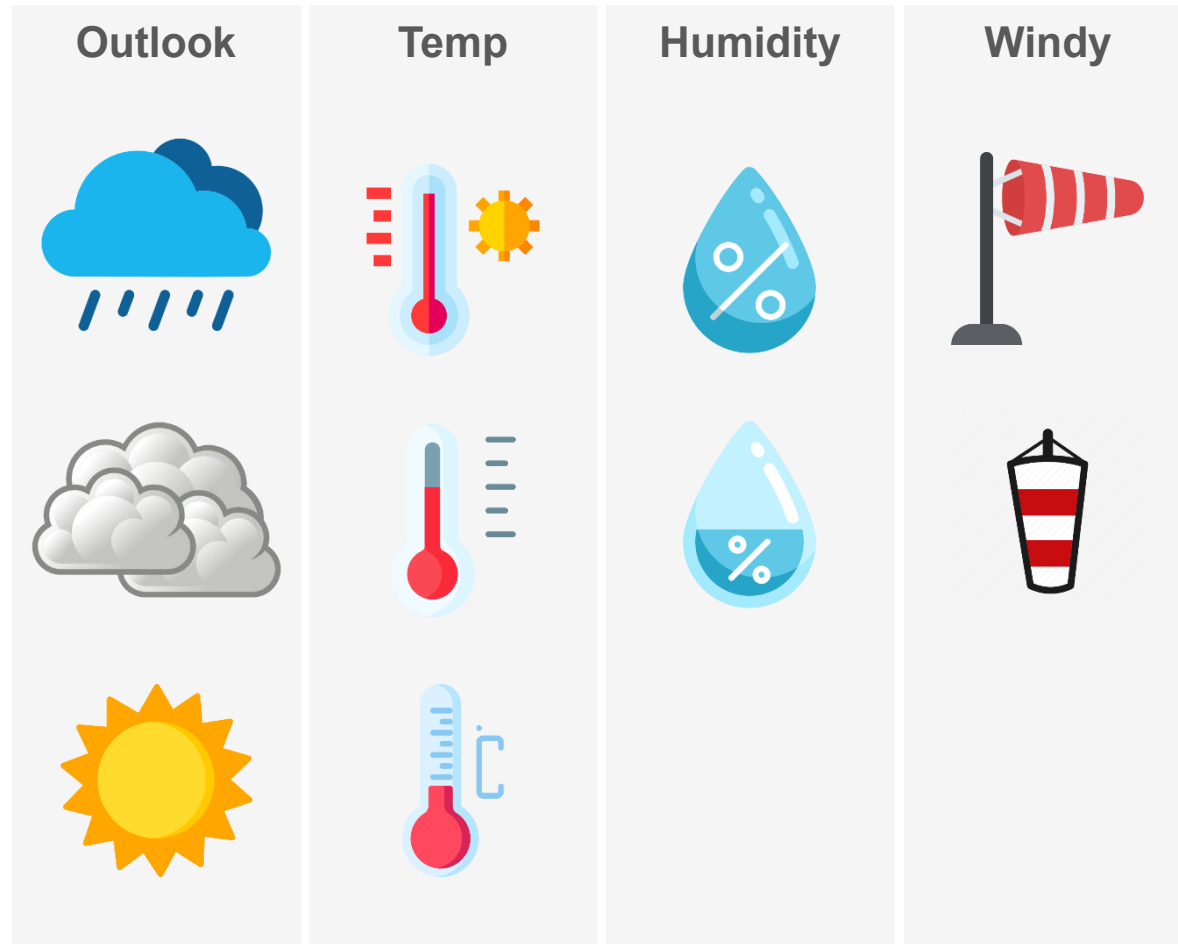
1. ZeroR

Simplicity First

- Simply provides for the category **based on the most frequent class**
- **No predictability power**, but is a **useful baseline** performance to other classification methods

1. ZeroR

The case Study



1. ZeroR

Example

Predictors

Target

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No



Play Golf	
Yes	No
9	5

Play Golf = “Yes” is the ZeroR model for our dataset with an accuracy of 0.64 (9/14)

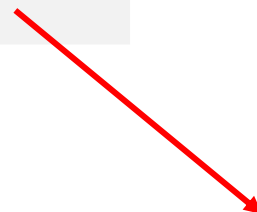
1. ZeroR

Example

Play Golf = “Yes” is the ZeroR model for our dataset

In a new observation (test set)...

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?



Yes

02. What is OneR?

A simple classification algorithm that **generates one rule for each predictor** in the data

2. OneR

Introduction

Learns a 1-level decision tree

- i.e. definition of rules where all of them test one particular attribute only

Basic Version

- One branch for each value
- Each branch assigns most frequent class
- Error rate: proportion of instances that don't belong to the majority class of their corresponding branch
- Choose attribute with lowest error rate

(assumes nominal attributes)

Note: missing is treated as a separate attribute value

2.1. OneR

Pseudocode for OneR

For each attribute,

For each value of the attribute, make a rule as follows:

count how often each class appears

find the most frequent class

make the rule assign that class to this attribute-value

Calculate the error rate of the rules

Choose the rules with the smallest error rate

2.2. OneR

Example

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

		Play	
		Yes	No
Outlook	Sunny	2	3
	Overcast	4	0
	Rainy	3	2

Attribute	Rules (most freq.)	Errors	Total Errors
Outlook	Sunny → No	2 / 5	4 / 14
	Overcast → Yes	0 / 4	
	Rainy → Yes	2 / 5	
Temp	Hot → No	2 / 4	5 / 14
	Mild → Yes	2 / 6	
	Cool → Yes	1 / 4	
Humidity	High → No	3 / 7	4 / 14
	Normal → Yes	1 / 7	
Windy	False → Yes	2 / 8	5 / 14
	True → No	3 / 6	

2.2. OneR

Example

Attribute	Rules	Errors	Total Errors
Outlook	Sunny → No	2 / 5	4 / 14
	Overcast → Yes	0 / 4	
	Rainy → Yes	2 / 5	
Temp	Hot → No	2 / 4	5 / 14
	Mild → Yes	2 / 6	
	Cool → Yes	1 / 4	
Humidity	High → No	3 / 7	4 / 14
	Normal → Yes	1 / 7	
Windy	False → Yes	2 / 8	5 / 14
	True → No	3 / 6	

		Play	
		Yes	No
Outlook	Sunny	2	3
	Overcast	4	0
	Rainy	3	2

If outlook = sunny then Play = No

If outlook = overcast then Play = Yes

If outlook = Rainy then Play = Yes

2.2. OneR

Example

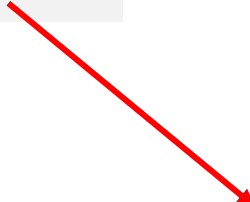
If outlook = sunny then Play = No

If outlook = overcast then Play = Yes

If outlook = Rainy then Play = Yes

In a new observation (test set)...

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?



No

03. What are Bayesian Classifiers?

A family of simple "probabilistic classifiers" based on applying Bayes' theorem with strong independence assumptions between the features

3. Bayesian Classifiers

Introduction to Bayesian Classifiers

A statistical classifier

- Performs probabilistic prediction, i.e., predict class membership probabilities

Foundation

- Based on Bayes theorem

Performance

- A simple Bayesian classifier, naïve Bayesian classifier, has comparable performance with decision tree and selected neural network classifiers

Incremental

- Each training example can incrementally increase / decrease the probability that a hypothesis is correct – prior knowledge can be combined with observed data

3. Bayesian Classifiers

Bayes Theorem

Probability of A and B occurring:

$$P(A \text{ and } B) = P(A) \times P(B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Then:

$$P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}$$

3. Bayesian Classifiers

Bayes Theorem

Probability of a hypothesis H is given that I have seen some evidence E

If we see some new evidence, we can update the believe on that hypothesis based on:

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

Where:

- $P(H)$ - what we believed before we saw the evidence
- $P(E|H)$ - the likelihood of seen that evidence if the hypothesis was correct
- $P(E)$ - the likelihood of that evidence under any circumstances

3. Naïve Bayes

Cookies Example

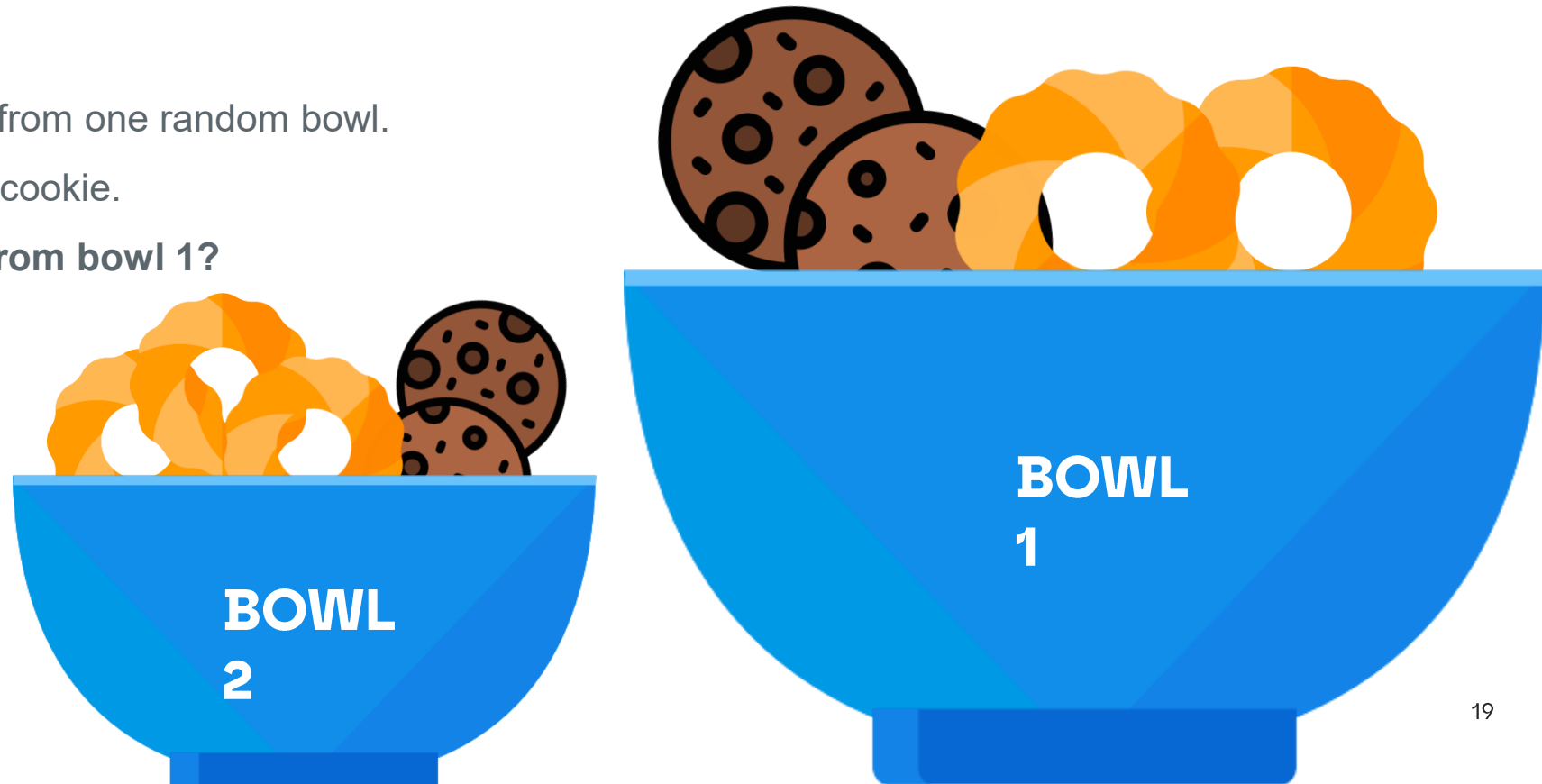
Suppose we have two bowls of cookies:

- Bowl 1 has 10 chocolate cookies and 30 butter cookies
- Bowl 2 has 20 chocolate cookies and 20 butter cookies

I just picked one cookie at random from one random bowl.

The cookie turns out to be a butter cookie.

How probable was that I took it from bowl 1?



3. Naïve Bayes

Cookies Example

Bowl 1 has 10 chocolate cookies and 30 butter cookies

Bowl 2 has 20 chocolate cookies and 20 butter cookies



H_1 cookie from bowl 1

H_2 cookie from bowl 2



$$P(H_1) = P(H_2) = \frac{1}{2}$$

$$P(E|H_1) = \frac{3}{4}$$

$$P(E|H_2) = \frac{1}{2}$$

$$P(E) = \frac{50}{80} = \frac{5}{8}$$

$$P(H_1|E) = \frac{P(H_1) \times P(E|H_1)}{P(E)} = \frac{\frac{1}{2} \times \frac{3}{4}}{\frac{5}{8}} = 0.6$$



3. Naive Bayes

Baye's Theorem Basics

Let E be a data sample

- E is considered an “evidence” with n attributes and class label unknown
- E is a customer described by age and salary

Let H be a hypothesis that E belong to class C

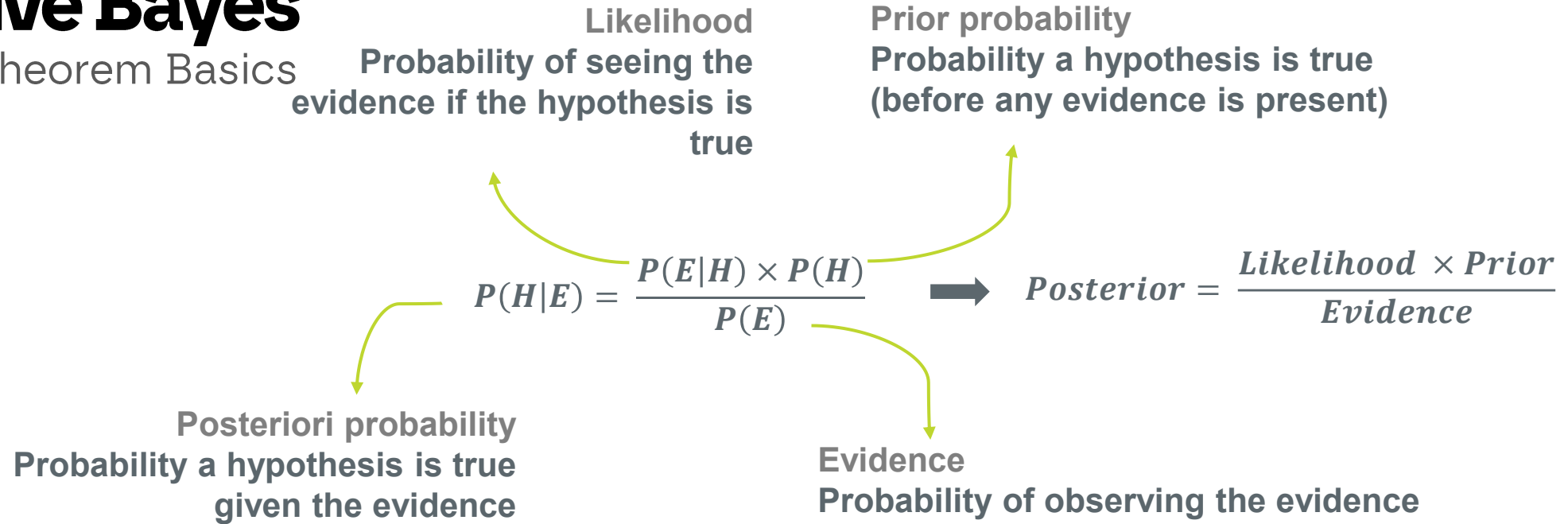
- Assume C is the class of computer buyers

In classification we want to determine $P(H|E)$

- The probability that the hypothesis holds given the observed data sample E
- Or the probability that E belongs to the class C given that we know the attributes of E

3. Naïve Bayes

Baye's Theorem Basics



Example

$P(H|E)$ probability that a customer will buy a computer given that he has 35 years old and has €4000 of income (What we want to know !)

$P(E|H)$ probability that a customer, E, is 35 years old and has €4000 of income, given that we know he bought a computer

$P(H)$ probability of any given customer to buy a computer, regardless of age and income

$P(E)$ probability that a person from our set of customers, has 35 years old and €4000 of income

03. What is Naïve Bayes?

Naïve Bayes is a classification technique based on Bayes' Theorem with an **assumption of independence among predictors**.

3.2. Bayesian Classifiers

Naive Bayes

The Naive Bayes classifier is based on two essential assumptions:

(i) Conditional Independence

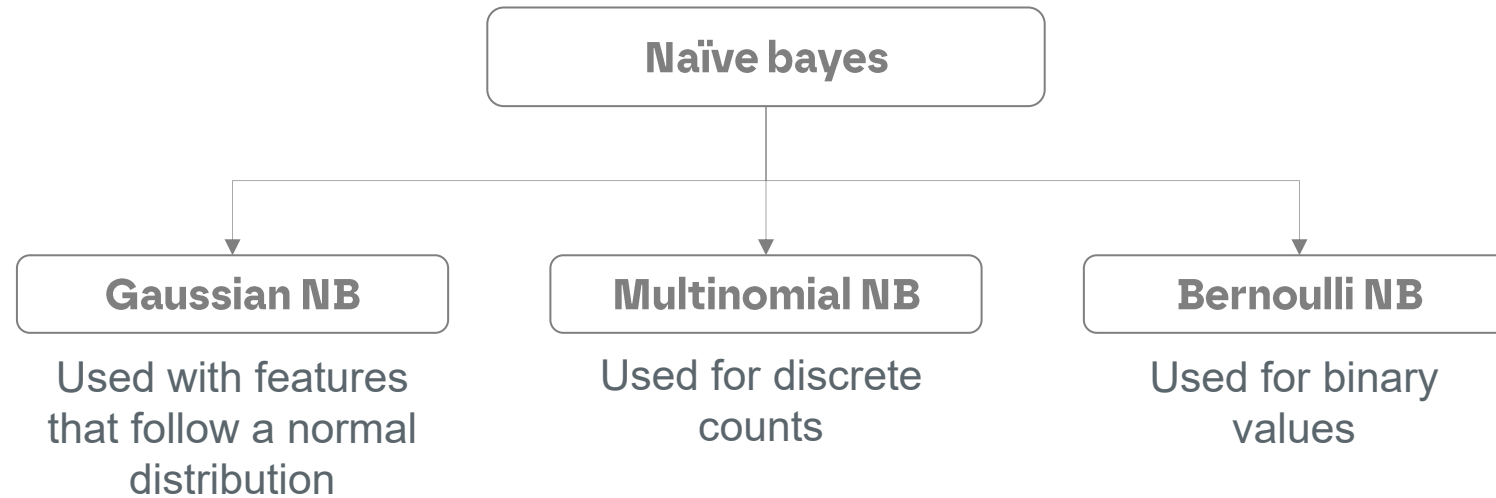
All features are independent of each other. This implies that one feature does not affect the performance of the other. This is the sole reason behind the 'Naïve' in 'Naïve Bayes.'

(ii) Feature Importance

All features are equally important. It is essential to know all the features to make good predictions and get the most accurate results.

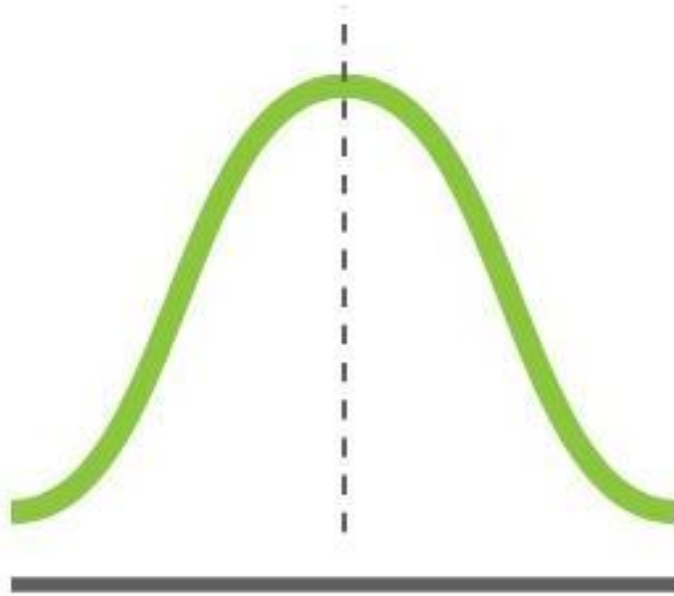
3.2. Bayesian Classifiers

Naive Bayes



3.2.1. Bayesian Classifiers

Gaussian Naïve Bayes



Gaussian
Naïve bayes

3.2.1. Bayesian Classifiers

Gaussian Naive Bayes

Height	Weight	Foot Size	Gender (T)
6	180	12	0
5.92	190	11	0
5.58	170	12	0
5.92	165	10	0
5	100	6	1
5.5	150	8	1
5.42	130	7	1
5.75	150	9	1

1. Calculate the class probability, i.e., the probability being male (0) or female (1)

$$P(\text{male}) = 4/8$$

$$P(\text{female}) = 4/8$$

In a new observation (test set)...

Height	Weight	Foot Size	Gender (T)
6	130	8	?

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

3.2.1. Bayesian Classifiers

Gaussian Naive Bayes

2. Calculate the individual probabilities for each feature

$$P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

Height	Weight	Foot Size	T
6	130	8	?

	Height	Weight	Foot Size	T
	6	180	12	0
	5.92	190	11	0
	5.58	170	12	0
	5.92	165	10	0
μ	5.855	176.250	11.250	
σ^2	0.035	122.916	0.917	

	Height	Weight	Foot Size	T
	5	100	6	1
	5.5	150	8	1
	5.42	130	7	1
	5.75	150	9	1
μ	5.418	132.5	7.5	
σ^2	0.097	558.33	1.667	

$$P(\text{height}|\text{male}) = \frac{1}{\sqrt{2\pi \times 0.0350}} \exp\left(-\frac{(6 - 5.855)^2}{2 \times 0.0350}\right) = \frac{1}{0.468947} \exp(-0.30036) = 1.579$$

$$P(\text{weight}|\text{male}) = 5.98 \times 10^{-6}$$

$$P(\text{footsize}|\text{male}) = 1.3112 \times 10^{-3}$$

$$P(\text{height}|\text{female}) = 0.22346$$

$$P(\text{weight}|\text{female}) = 1.679 \times 10^{-2}$$

$$P(\text{footsize}|\text{female}) = 0.28669$$

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

3.2.1. Bayesian Classifiers

Gaussian Naive Bayes

3. Find the probabilities of being male or female

Height	Weight	Foot Size	T
6	180	12	0
5.92	190	11	0
5.58	170	12	0
5.92	165	10	0
5	100	6	1
5.5	150	8	1
5.42	130	7	1
5.75	150	9	1

x	6	130	8	?
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male x	1.579	5.98×10^{-6}	1.31×10^{-3}
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female x	0.223	1.68×10^{-2}	0.287
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$$P(\text{male}|x) = \frac{0.5 * 1.579 * 5.98 \times 10^{-6} * 1.31 \times 10^{-3}}{P(x) * } = \frac{6.1984 \times 10^{-9}}{P(x)}$$

$$P(\text{female}|x) = \frac{0.5 * 0.22346 * 1.6789 \times 10^{-2} * 0.28669}{P(x) * } = \frac{5.3778 \times 10^{-4}}{P(x)}$$

$$* P(x) = P(\text{male} | x) * P(\text{male}) + P(\text{female} | x) * P(\text{female})$$

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

3.2.1. Bayesian Classifiers

Gaussian Naive Bayes

In the new observation (test set)...

Height	Weight	FootSize	Gender
6	130	8	?

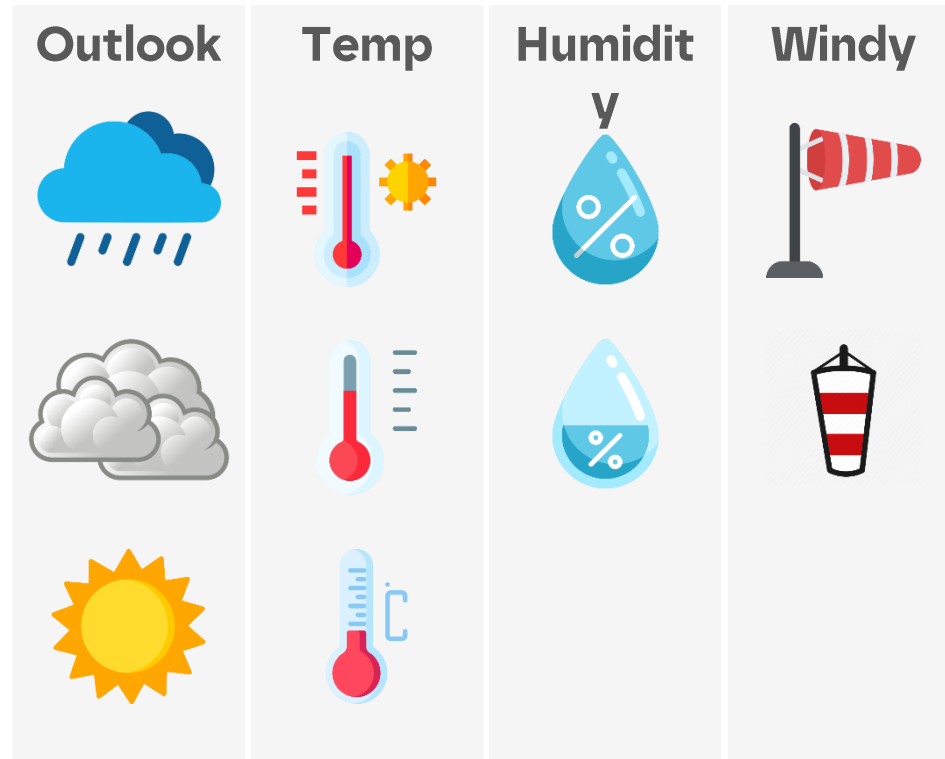

$$\begin{aligned}\text{Male} &= (6.1984 \times 10^{-6}) / P(x) \\ \text{Female} &= (5.3778 \times 10^{-4}) / P(x)\end{aligned}$$



Female is Higher, so Female

3.2.2. Bayesian Classifiers

Multinomial Naive Bayes



Multinomial
Naïve bayes

3.2.2. Bayesian Classifiers

Multinomial Naive Bayes

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?

3.2.2. Bayesian Classifiers

Multinomial Naive Bayes

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?

$$P(\text{Yes}|E) = P(\text{Outlook} = \text{Sunny}|\text{yes}) \times P(\text{Temp} = \text{Cool}|\text{yes}) \times P(\text{Humidity} = \text{High}|\text{yes}) \times P(\text{Windy} = \text{True}|\text{yes}) \times P(\text{Yes})$$

Naïve Bayes Classifier

- We assume the attributes are conditionally independent (i.e.: no dependence relation between attributes)
- This greatly reduces the computation costs: Only count the class distribution

3.2. Bayesian Classifiers

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Outlook			Temperature			Humidity			Windy			Play	
	Yes	No		Yes	No		Yes	No		Yes	No	Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2	9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3		
Rainy	3	2	Cool	3	1								
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5	9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5		
Rainy	3/9	2/5	Cool	3/9	1/5								

Likelihood of the two classes

- For “Yes” = $2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.0053$
- For “No” = $3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.0206$

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?

Conversion into a probability by normalization:

$$P(\text{Yes}) = \frac{0.0053}{0.0053 + 0.0206} = 0.205$$

3.2.2. Bayesian Classifiers

Multinomial Naive Bayes

In a new observation (test set)...

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?

Yes = 0.205

No = 0.795



No

3.2.2. Bayesian Classifiers

Multinomial Naive Bayes – Laplace Smoothing

Now let's imagine that my new observation is something like this:

Outlook	Temp	Humidity	Windy	Play Golf
Snow	Cool	High	True	?

We don't have any snow in the train dataset...

And if we apply the formula:

Likelihood of the two classes

- For "Yes" = $0/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0$
- For "No" = $0/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0$

This is the problem of zero probability. So, how to deal with this problem?



Laplace Smoothing

3.2. Bayesian Classifiers

Naive Bayes

How can we solve this?

Just add +1 to each possible value in that feature....

Outlook		
	Yes	No
Sunny	2	3
Overcast	4	0
Rainy	3	2
Sunny	2/9	3/5
Overcast	4/9	0/5
Rainy	3/9	2/5



Outlook		
	Yes	No
Sunny	3	4
Overcast	5	1
Rainy	4	3
Snow	1	1
Sunny	3/13	4/9
Overcast	5/13	1/9
Rainy	4/13	3/9
Snow	1/13	1/9



Likelihood of the two classes

For "Yes" = $1/13 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.00549$

For "No" = $1/9 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.00381$

Conversion into a probability by normalization:

$$P(\text{Yes}) = \frac{0.00549}{0.00549 + 0.00381} = 0.59$$

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

0	1	1	0
1	0	0	1
1	0	1	0

Bernoulli
Naïve bayes

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

- Bernoulli Naive Bayes is a part of the Naive Bayes family.
- It is based on the Bernoulli Distribution and accepts only binary values, i.e., 0 or 1.
- If the features of the dataset are binary, then we can assume that Bernoulli Naive Bayes is the algorithm to be used.

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

Gender	Cough	Fever	Disease
Female	Yes	No	False
Female	Yes	Yes	True
Male	No	Yes	False
Male	No	No	True
Male	Yes	Yes	True

Gender	Cough	Fever	Disease
Male	Yes	Yes	?

1. Calculate the class probability, i.e., the probability of disease or not

$$P(\text{Disease} = \text{True}) = 3/5$$

$$P(\text{Disease} = \text{False}) = 2/5$$

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

Gender	Cough	Fever	Disease
Female	Yes	No	False
Female	Yes	Yes	True
Male	No	Yes	False
Male	No	No	True
Male	Yes	Yes	True
Gender	Cough	Fever	Disease
Male	Yes	Yes	?

2. Calculate the individual probabilities for each feature

TRUE

Gender	Cough	Fever	Disease
Female	Yes	Yes	True
Male	No	No	True
Male	Yes	Yes	True

$$P(\text{Gender} = \text{Male} \mid \text{Disease} = \text{True}) = \frac{2}{3}$$

$$P(\text{Cough} = \text{Yes} \mid \text{Disease} = \text{True}) = \frac{2}{3}$$

$$P(\text{Fever} = \text{Yes} \mid \text{Disease} = \text{True}) = \frac{2}{3}$$

FALSE

Gender	Cough	Fever	Disease
Female	Yes	No	False
Male	No	Yes	False

$$P(\text{Gender} = \text{Male} \mid \text{Disease} = \text{False}) = \frac{1}{2}$$

$$P(\text{Cough} = \text{Yes} \mid \text{Disease} = \text{False}) = \frac{1}{2}$$

$$P(\text{Fever} = \text{Yes} \mid \text{Disease} = \text{False}) = \frac{1}{2}$$

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

Gender	Cough	Fever	Disease
Female	Yes	No	False
Female	Yes	Yes	True
Male	No	Yes	False
Male	No	No	True
Male	Yes	Yes	True
Gender	Cough	Fever	Disease
Male	Yes	Yes	?

$$P(\text{Gender} = \text{Male} \mid \text{Disease} = \text{True}) = \frac{2}{3}$$

$$P(\text{Cough} = \text{Yes} \mid \text{Disease} = \text{True}) = \frac{2}{3}$$

$$P(\text{Fever} = \text{Yes} \mid \text{Disease} = \text{True}) = \frac{2}{3}$$

$$P(\text{Gender} = \text{Male} \mid \text{Disease} = \text{False}) = \frac{1}{2}$$

$$P(\text{Cough} = \text{Yes} \mid \text{Disease} = \text{False}) = \frac{1}{2}$$

$$P(\text{Fever} = \text{Yes} \mid \text{Disease} = \text{False}) = \frac{1}{2}$$

3. Find the probabilities of having or not the disease

TRUE

$$P(\text{Disease} = \text{True} \mid x) = \frac{P(x \mid \text{Disease} = \text{True}) \times P(\text{Disease} = \text{True})}{P(x)} = \frac{\left(\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}\right) \times \left(\frac{3}{5}\right)}{P(x)} = \frac{0.17}{P(x)}$$

FALSE

$$P(\text{Disease} = \text{False} \mid x) = \frac{P(x \mid \text{Disease} = \text{False}) \times P(\text{Disease} = \text{False})}{P(x)} = \frac{\left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right) \times \left(\frac{2}{5}\right)}{P(x)} = \frac{0.05}{P(x)}$$

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

Gender	Cough	Fever	Disease
Female	Yes	No	False
Female	Yes	Yes	True
Male	No	Yes	False
Male	No	No	True
Male	Yes	Yes	True
Gender	Cough	Fever	Disease
Male	Yes	Yes	?

4. Calculate the estimator probability

$$P(x) = P(\text{Gender} = \text{Male}) \times P(\text{Cough} = \text{Yes}) \times P(\text{Fever} = \text{Yes}) = \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} = \frac{27}{125} = 0.21$$

3.2.3. Bayesian Classifiers

Bernoulli Naive Bayes

Gender	Cough	Fever	Disease
Female	Yes	No	False
Female	Yes	Yes	True
Male	No	Yes	False
Male	No	No	True
Male	Yes	Yes	True
Gender	Cough	Fever	Disease
Male	Yes	Yes	?

Finally:

$$P(\text{Disease} = \text{True} \mid x) = \frac{0.17}{P(x)} = \frac{0.17}{0.21} = 0.80 \quad \textcircled{1}$$

$$P(\text{Disease} = \text{False} \mid x) = \frac{0.05}{P(x)} = \frac{0.05}{0.21} = 0.23 \quad \textcircled{2}$$

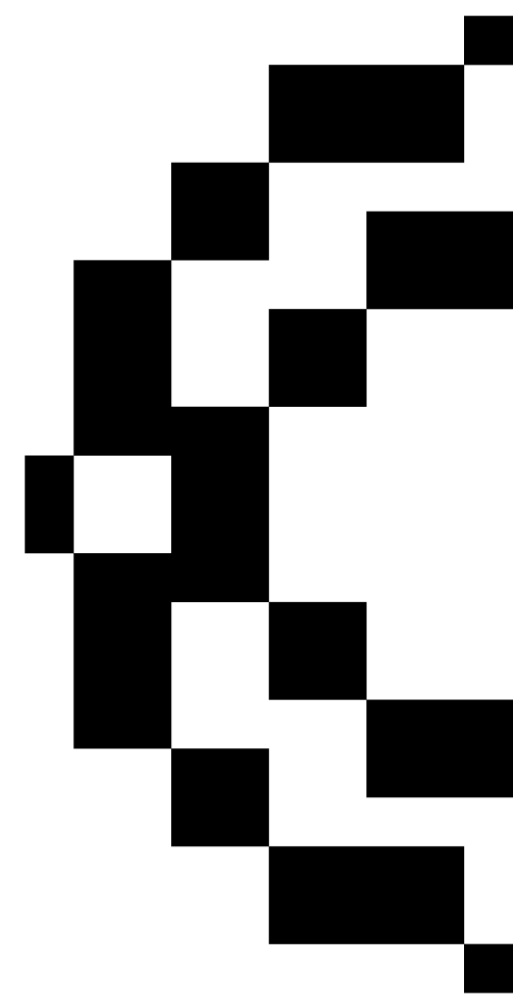
Notice that $\textcircled{1} > \textcircled{2}$, so the result of the instance x is True, i.e., the person has the disease.

3.2. Bayesian Classifiers

Naive Bayes – Advantages and Disadvantages

-  This algorithm works quickly and can save a lot of time.
-  Naive Bayes is suitable for solving multi-class prediction problems.
-  If its assumption of the independence of features holds true, it can perform better than other models and requires much less training data.
-  Naive Bayes is better suited for categorical input variables than numerical variables.
-  Naive Bayes assumes that all predictors (or features) are independent, rarely happening in real life. This limits the applicability of this algorithm in real-world use cases.
-  This algorithm faces the 'zero-frequency problem' where it assigns zero probability to a categorical variable whose category in the test data set wasn't available in the training dataset. It would be best if you used a smoothing technique to overcome this issue.
-  Its estimations can be wrong in some cases, so you shouldn't take its probability outputs very seriously.

04. What are Instance Based Classifiers?



4. Instance Based Classifiers

Introduction

Simplest form of learning (rote learning)

- Training instances are searched for instance that most closely resembles new instance
- The instances themselves represent the knowledge
- Also called instance-based learning

Instance based learning is lazy learning (vs eager learning)

Methods

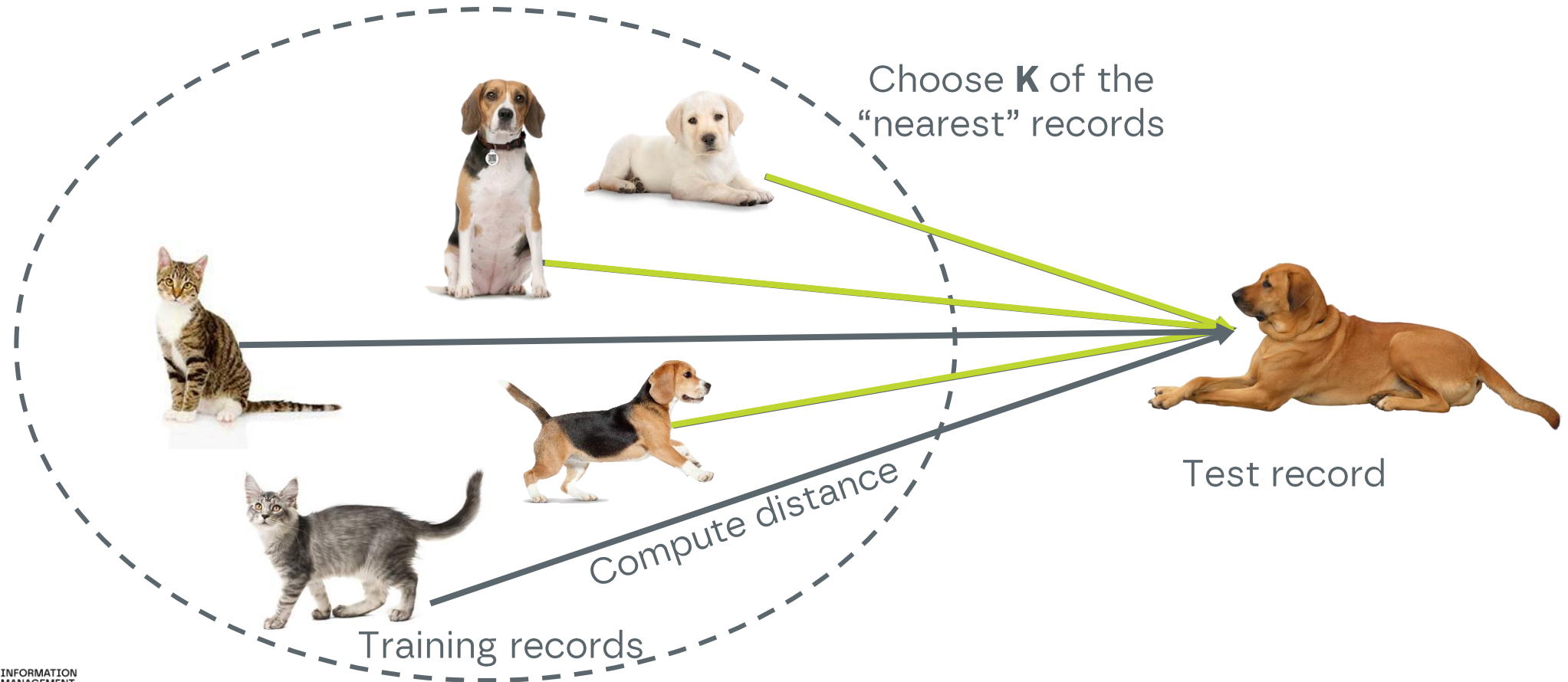
- Nearest neighbor
- K-nearest neighbor
- ...

4. Instance Based Classifiers

Introduction

Basic Idea

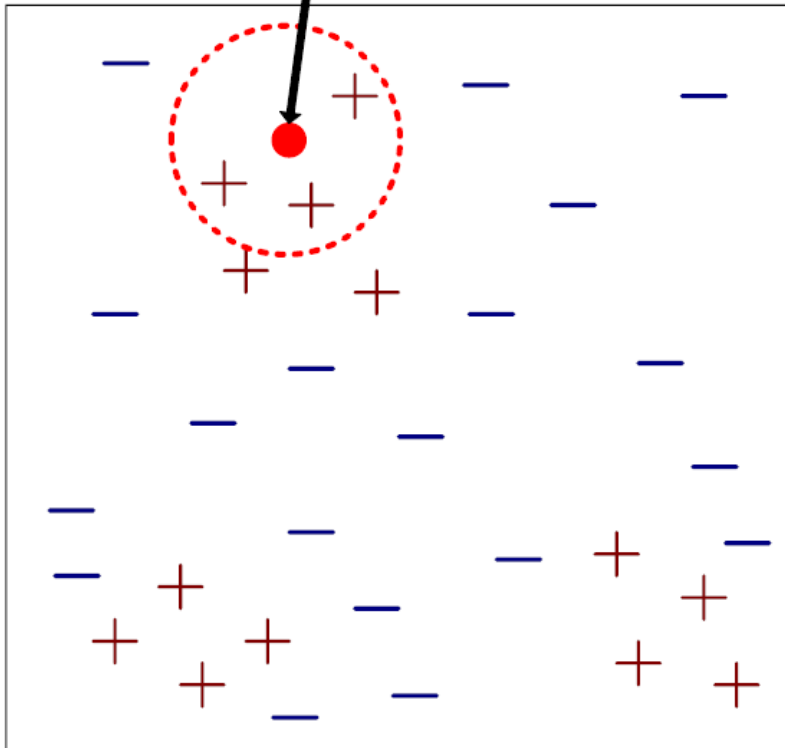
- If it walks like a dog, barks like a dog, then it's probably a dog



4.1. Instance Based Classifiers

Nearest Neighbor Classifier

Unknown record



Requires three things:

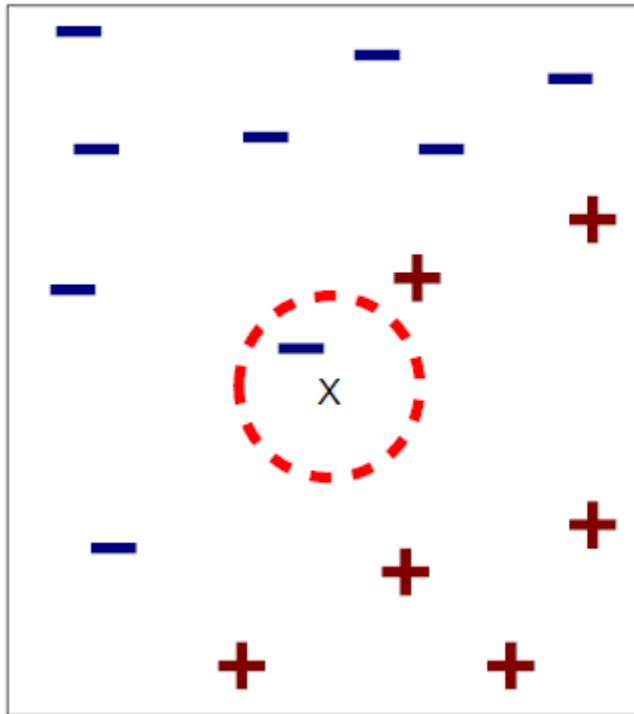
1. The set of stored records
2. Distance Metric to compute distance between records
3. The value of k , the number of nearest neighbors to retrieve

To classify an unknown record:

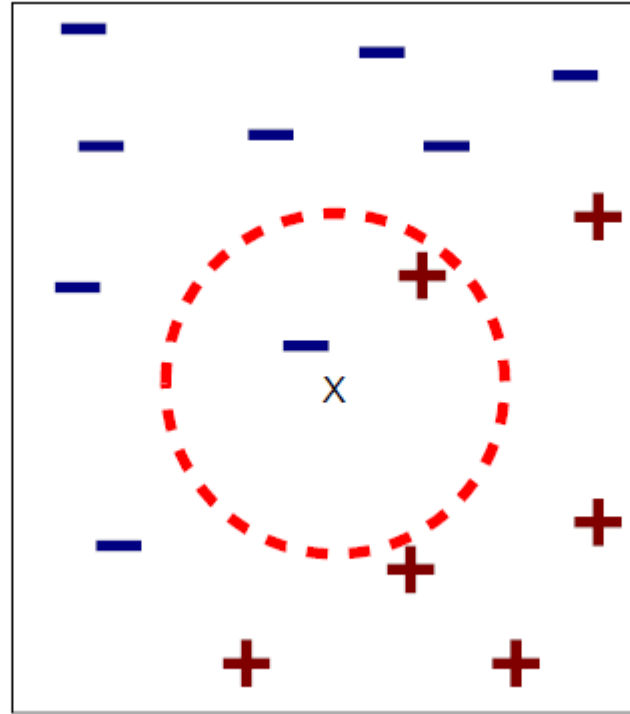
- Compute distance to other training records
- Identify k -nearest neighbors
- Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

4.1. Instance Based Classifiers

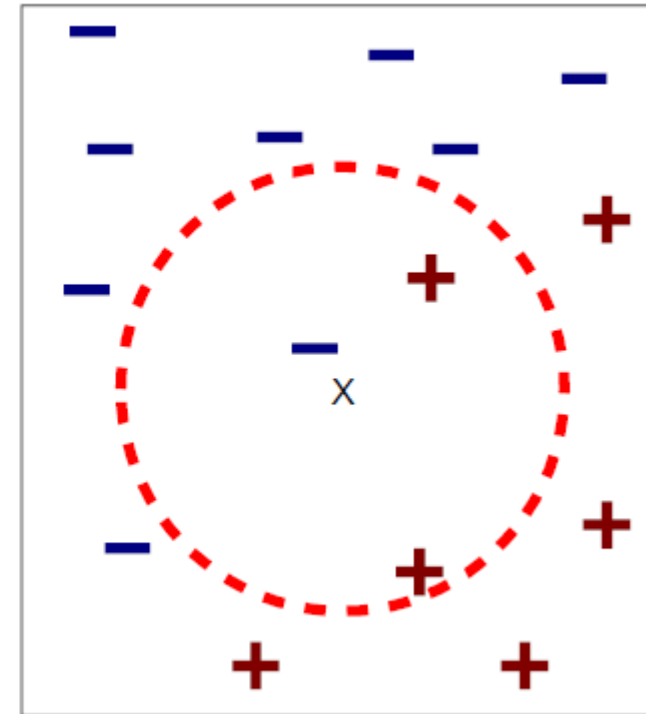
Nearest Neighbor Classifier



1 nearest neighbor



2 nearest neighbors



3 nearest neighbors

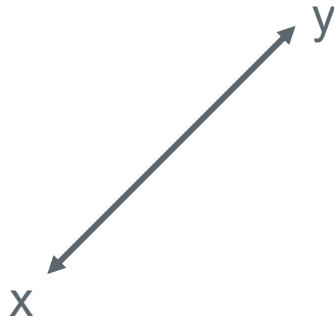
K-nearest neighbors of a record x are data points that have the k smallest distance to x

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

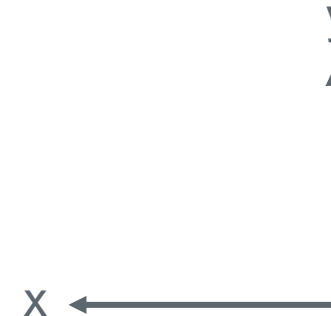
1. Compute distance between two points:

Euclidean distance



$$d(x, y) = \sqrt{\sum_{k=1}^p (x_k - y_k)^2}$$

Manhattan distance



$$d(x, y) = \sum_{k=1}^p |x_k - y_k|$$

Use it when dealing with high dimensionality

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

2. Determine the class from nearest neighbor list

- take the majority vote of class labels among the k-nearest neighbors
- Weigh the vote according to distance
 - weight factor, $W = \frac{1}{d^2}$

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

Choosing the value of k :

If k is too small:

sensitive to noise points

very sensitive to outliers

Overfitting

If k is too large:

neighborhood may include points from other classes

unable to detect small variations

Underfitting

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

Scaling issues

Attributes may have to be scaled to prevent distance measures from being dominated by one of the attributes

Example:

- height of a person may vary from 1.5m to 2m
- weight of a person may vary from 50Kg to 90Kg
- income of a person may vary from €5k to €500k

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

- k-NN classifiers are lazy learners
- It does not build models explicitly, unlike eager learners such as decision tree induction and rule-based systems
- Classifying unknown records are relatively expensive

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

- Have to calculate the distance of the test case from all training cases
- There may be irrelevant attributes amongst the attributes –curse of dimensionality
- Require a lot of memory to store the training set
- Require a LOT of time at the classification stage
- Very sensitive to outliers
- Very sensitive to the distance function chosen

4.1. Instance Based Classifiers

ID	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No



Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Cool	High	True	?

ID	Outlook	Temp	Humidity	Windy	Play Golf	Dist
1	0	1	0	1	No	2
2	0	1	0	0	No	1
3	1	1	0	1	Yes	3
4	1	1	0	1	Yes	3
5	1	0	1	1	Yes	3
6	1	0	1	0	No	2
7	1	0	1	0	Yes	2
8	0	1	0	1	No	2
9	0	0	1	1	Yes	2
10	1	1	1	1	Yes	4
11	0	1	1	0	Yes	2
12	1	1	0	0	Yes	2
13	1	0	1	1	Yes	3
14	1	0	0	0	No	1

4.1. Instance Based Classifiers

Nearest Neighbor Classifier Example

ID	Play Golf	Distance
1	No	2
2	No	1
3	Yes	3
4	Yes	3
5	Yes	3
6	No	2
7	Yes	2
8	No	2
9	Yes	2
10	Yes	4
11	Yes	2
12	Yes	2
13	Yes	3
14	No	1

In Sklearn:

- if it is found that two neighbors, neighbor $k+1$ and k , have identical distances but different labels, the results will depend on the ordering of the training data.

1 nearest neighbor (ID 2) $\rightarrow$ 1 “No” $\rightarrow$ “No”

2 nearest neighbors (ID 2 and 14) $\rightarrow$ 2 “No” $\rightarrow$ “No”

3 nearest neighbors (ID 2, 14 and 1) $\rightarrow$ 3 “No” $\rightarrow$ “No”

...

9 nearest neighbors (ID 2, 14, 1, 6, 7, 8, 9, 11, 12) $\rightarrow$ 5 “No” and 4 “Yes” $\rightarrow$ “No”

...

4.1. Instance Based Classifiers

Nearest Neighbor Classifier

My training data...

ID	Var1	Var2	Var3	Var4	Target
1	3	7	4	3	No
2	5	4	3	5	No
3	4	5	4	3	No
4	8	3	2	7	Yes
5	5	6	5	8	Yes
6	5	2	5	3	No
7	3	4	8	6	Yes
8	6	5	9	4	No

My test data

ID	Var1	Var2	Var3	Var4	Target
	3	6	4	5	?

ID	Dist Var1	Dist Var2	Dist Var3	Dist Var4	Dist Target	Total Dist
1	0	1	0	2	No	3
2	2	2	1	0	No	5
3	1	1	0	2	No	4
4	5	3	2	2	Yes	12
5	2	0	1	3	Yes	6
6	2	4	1	2	No	9
7	0	2	4	1	Yes	7
8	3	1	5	1	No	10

1 nearest neighbor → 1 “No” → “No”

2 nearest neighbors → 2 “No” → “No”

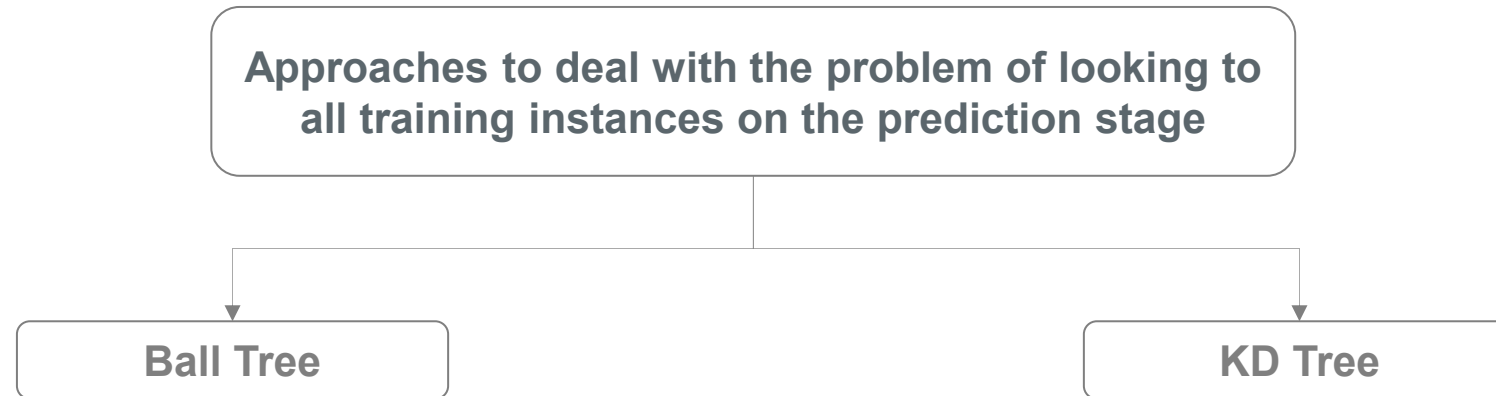
3 nearest neighbors → 3 “No” → “No”

4 nearest neighbors → 3 “No” and 1 “Yes” → “No”

...

4. Instance Based Classifiers

Additional Approaches to implement on KNN





Ball Tree Algorithm

4.2. Instance Based Classifiers

Ball Tree Algorithm

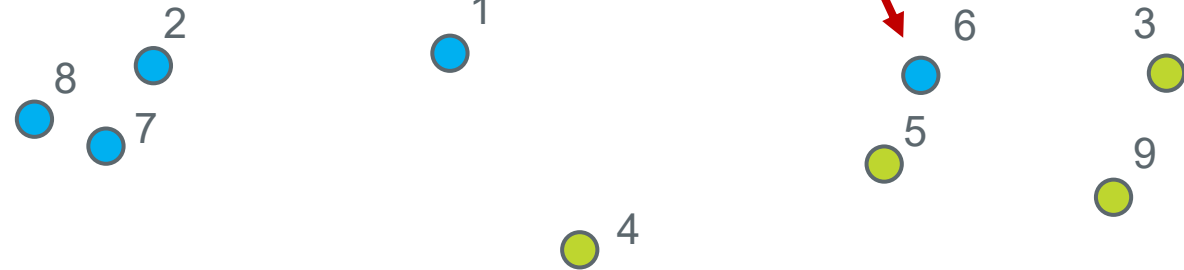
- Speed up the discovery of neighborhood points
- Ball Tree algorithm is a metric tree – organize and structure data points considering the metric space in which the points are located
- It has a hierarchical structure

4.2. Instance Based Classifiers

Ball Tree Algorithm

1ST STEP

Select one random data point (ID6)



4.2. Instance Based Classifiers

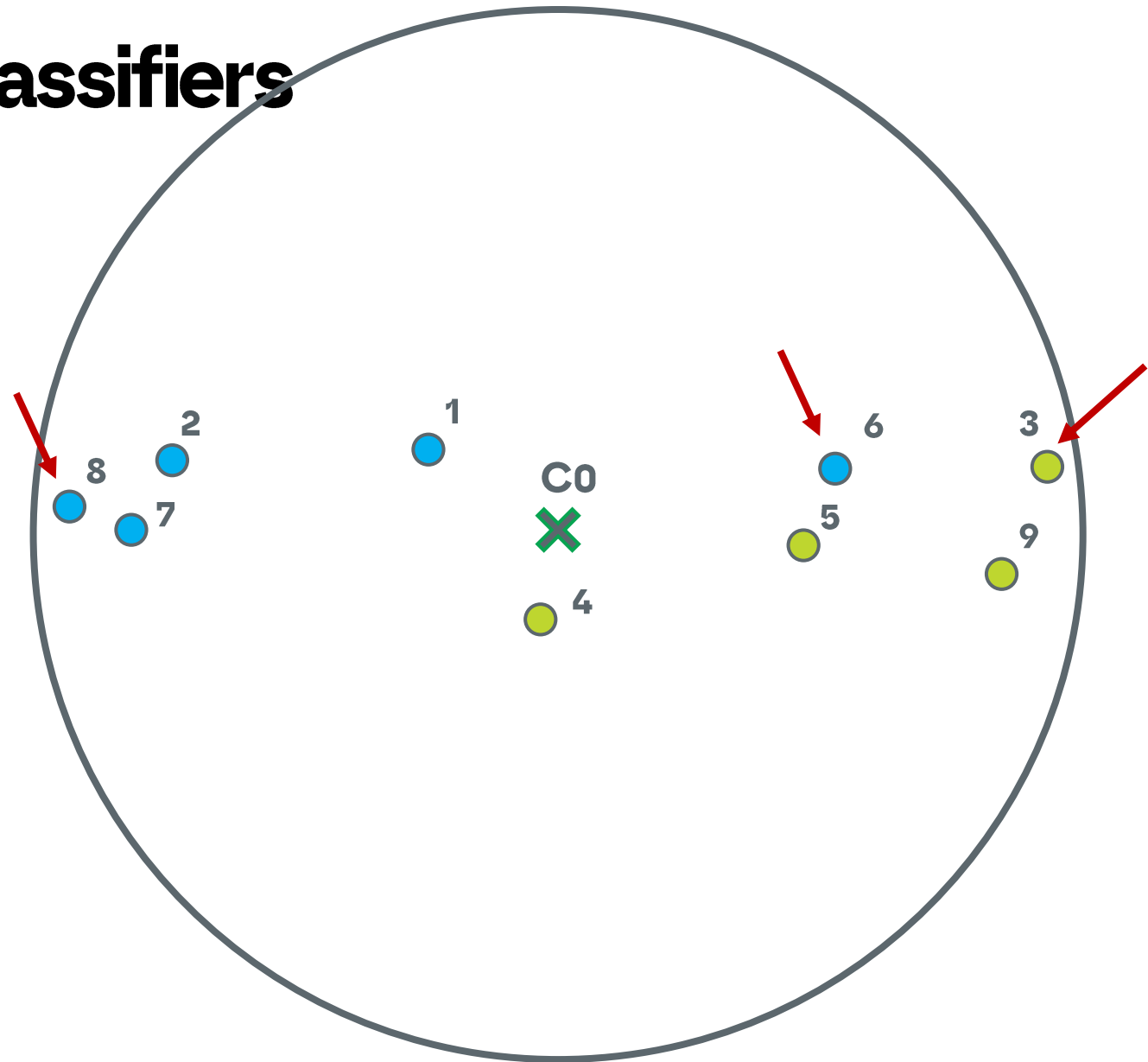
Ball Tree Algorithm

2ND STEP

Measure the distance between the selected data point (ID6) and all the other observations and get the one that is farthest (ID8).

Using the last selected observation (ID8), measure now the distance to all the other data points and get the one that is more distant (ID3).

Those are the limits for our first cluster, and the centroid is set (C0).



4.2. Instance Based Classifiers

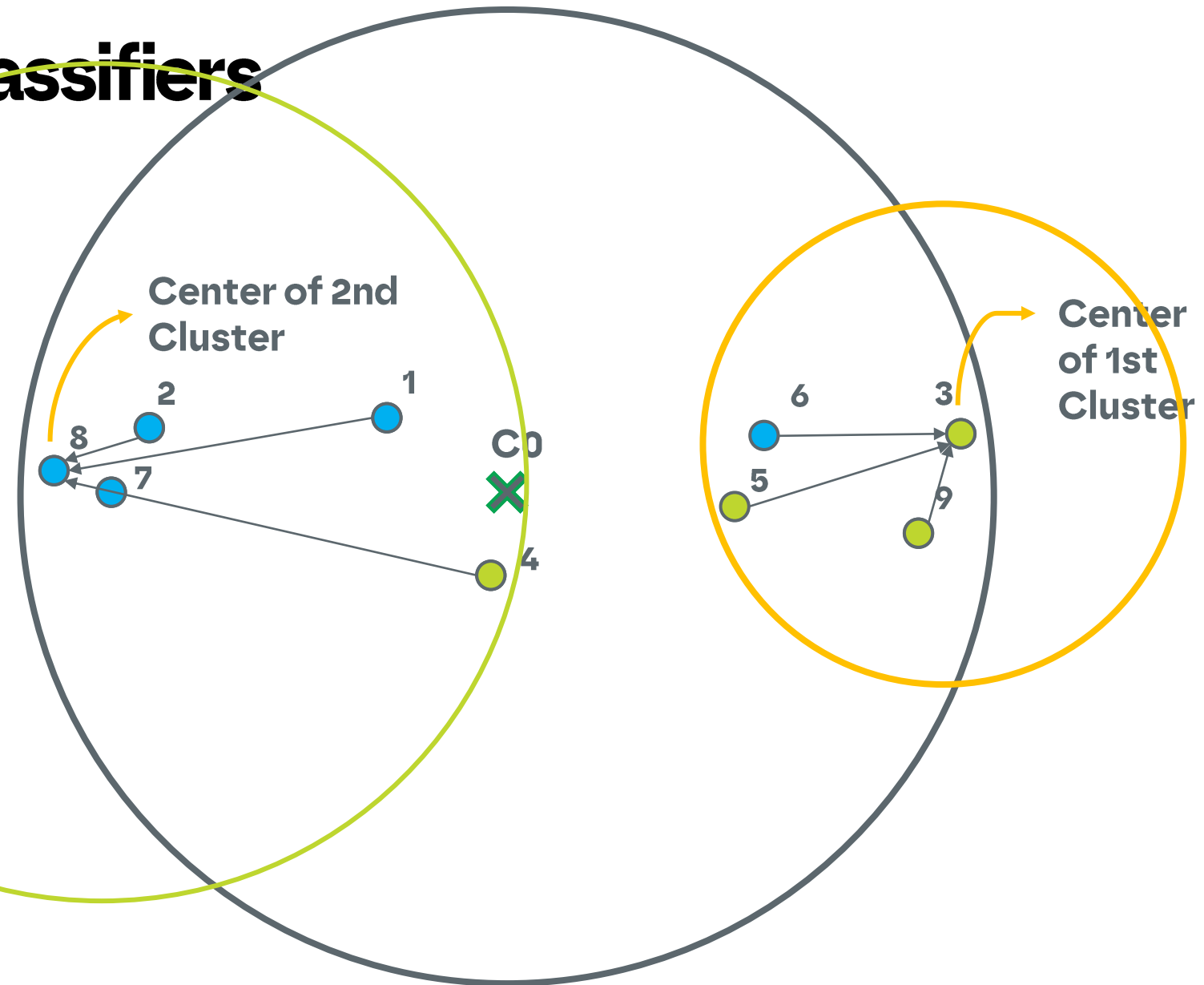
Ball Tree Algorithm

3RD STEP

The point with the maximum distance to the centroid is selected as the center of the first cluster and child node (orange cluster with ID3).

The point furthest away from the center of the first cluster is chosen as the center point of the second cluster (green cluster with ID8).

All other data points are assigned to the node and to the closest centroid.



4.2. Instance Based Classifiers

Ball Tree Algorithm

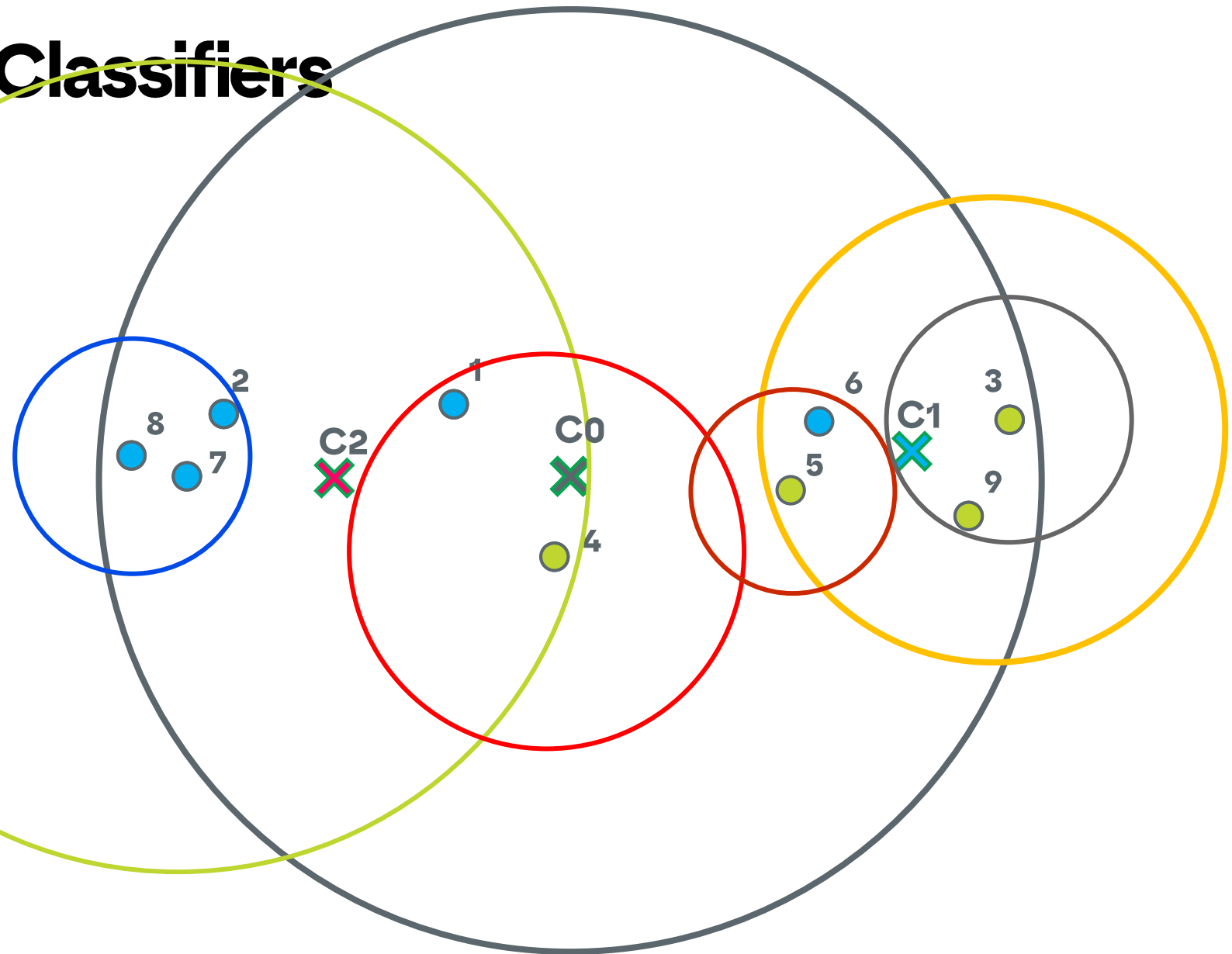
4th STEP

Taking into account the data points associated to each cluster, get the new centroids (C1 and C2).

The point with the maximum distance to the new centroid (for example, ID5 for C1) is selected as the center of the cluster and child node.

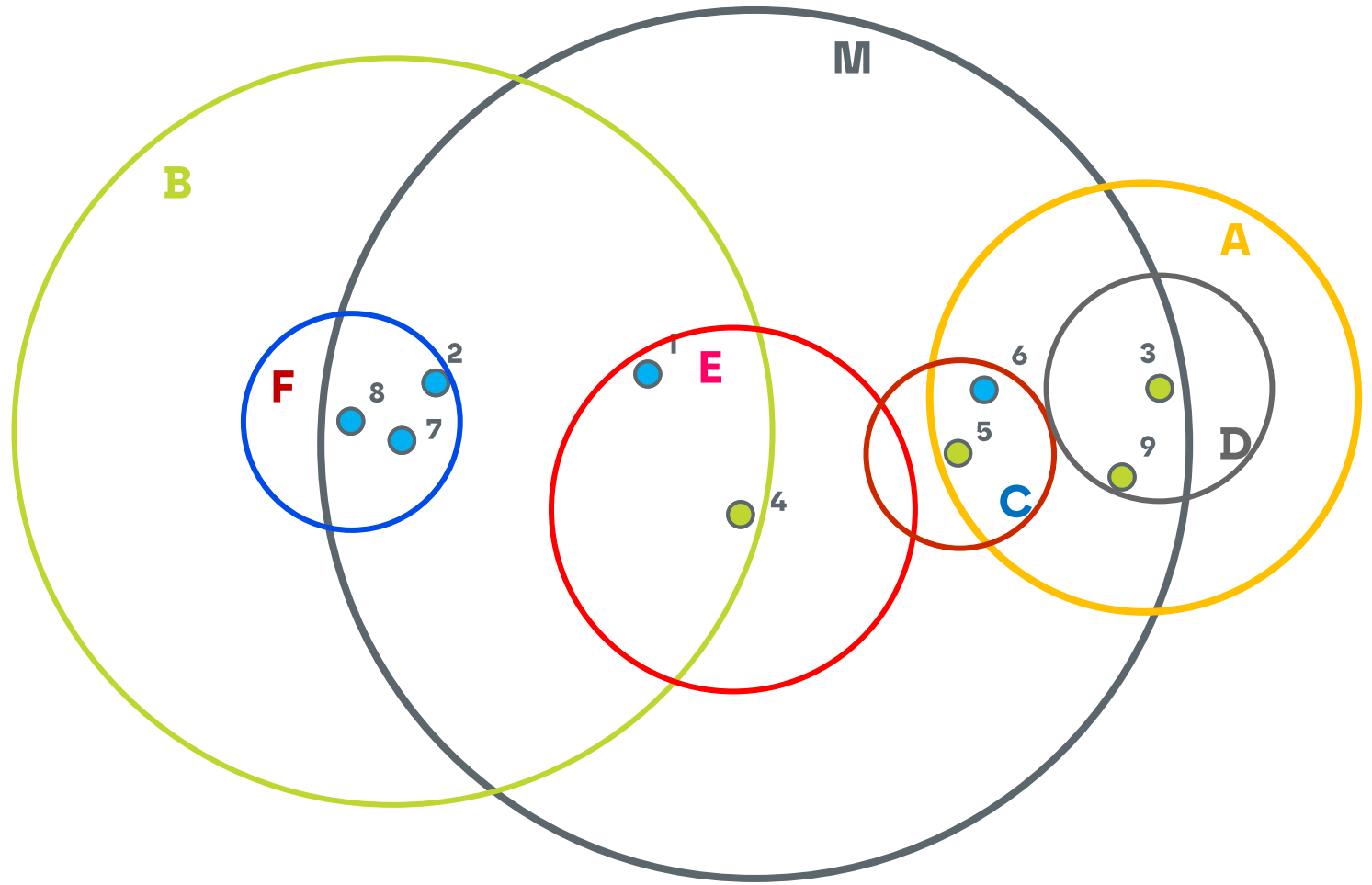
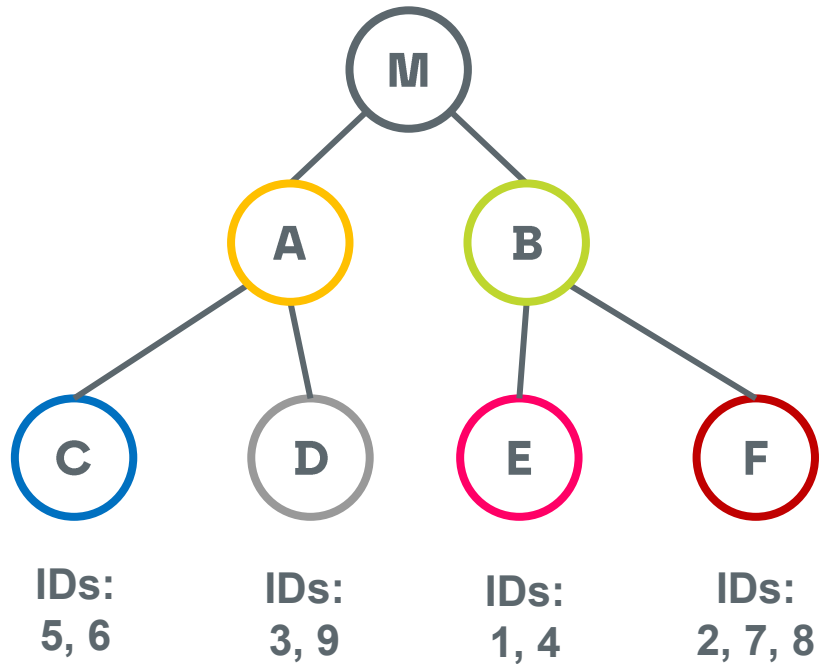
The point furthest away from the center of the cluster just defined is chosen as the center point of the next cluster (ID3).

Repeat until the desired depth.



4.2. Instance Based Classifiers

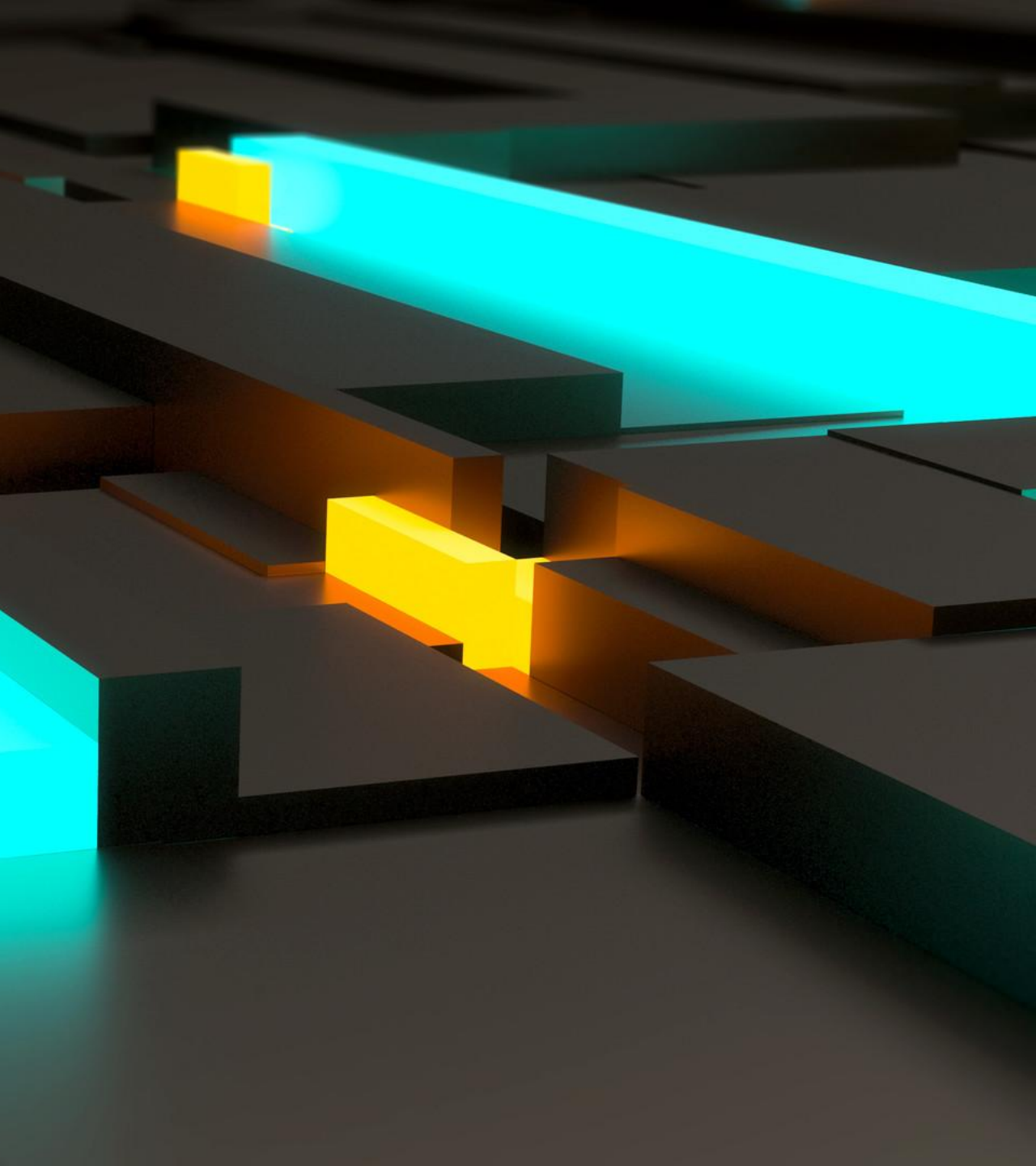
Ball Tree Algorithm



4.2. Instance Based Classifiers

Ball Tree Algorithm

- Works with clusters, where in 2-dimensions, each resemble to a ball, and in a multidimensional space, each “ball” is called a hypersphere
- Very efficient especially in situations when number of dimensions is very large.
- Any point in n-dimensional space will belong to either cluster but not to both – each point will belong to the cluster which centroid is closer
- Ball or hyperspheres may intersect but each data point will belong to only one cluster
- An unclassified (target) point must fall within any one of the nested balls. Points within this nested ball are expected to be nearest to target point.
- This accelerates the classification process.



KDTree Algorihtm

4.3. Instance Based Classifiers

KDTree Algorithm

- Speed up the discovery of neighborhood points
- It construct a sorted hierarchical data structure called k-dimensional Tree (KDTree)
- It is a binary tree
- The general idea of KDTrees is to partition the feature space.

4.3. Instance Based Classifiers

KDTree Algorithm

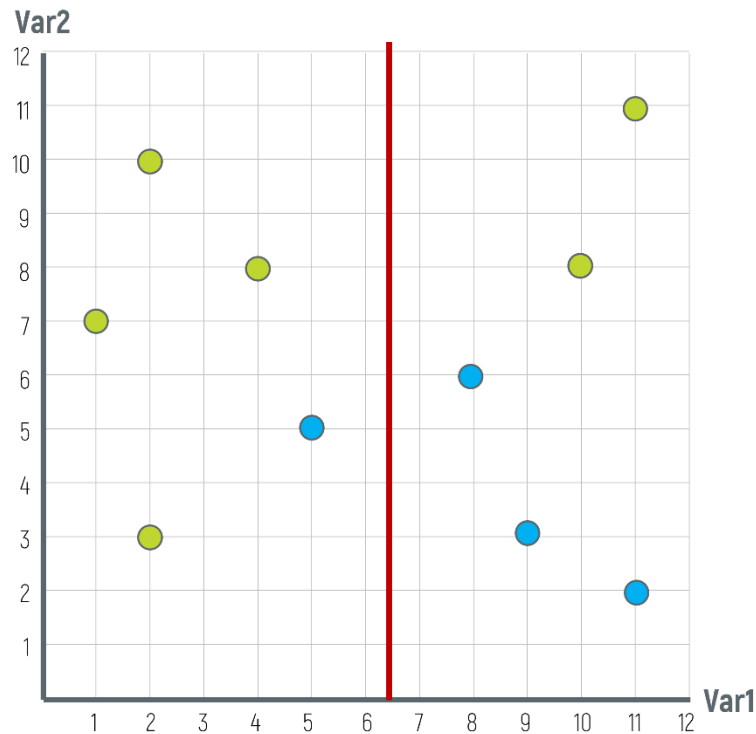
ID	Var1	Var2	Target
1	4	8	Green
2	9	3	Blue
3	11	11	Blue
4	11	2	Blue
5	1	7	Green
6	2	3	Green
7	5	5	Blue
8	2	10	Green
9	8	6	Blue
10	10	8	Green



4.3. Instance Based Classifiers

KDTree Algorithm

Step 1 – Start with first feature of the list, find the median for it and split the data accordingly (the datapoint corresponding to the median will be the indexed point of that node)



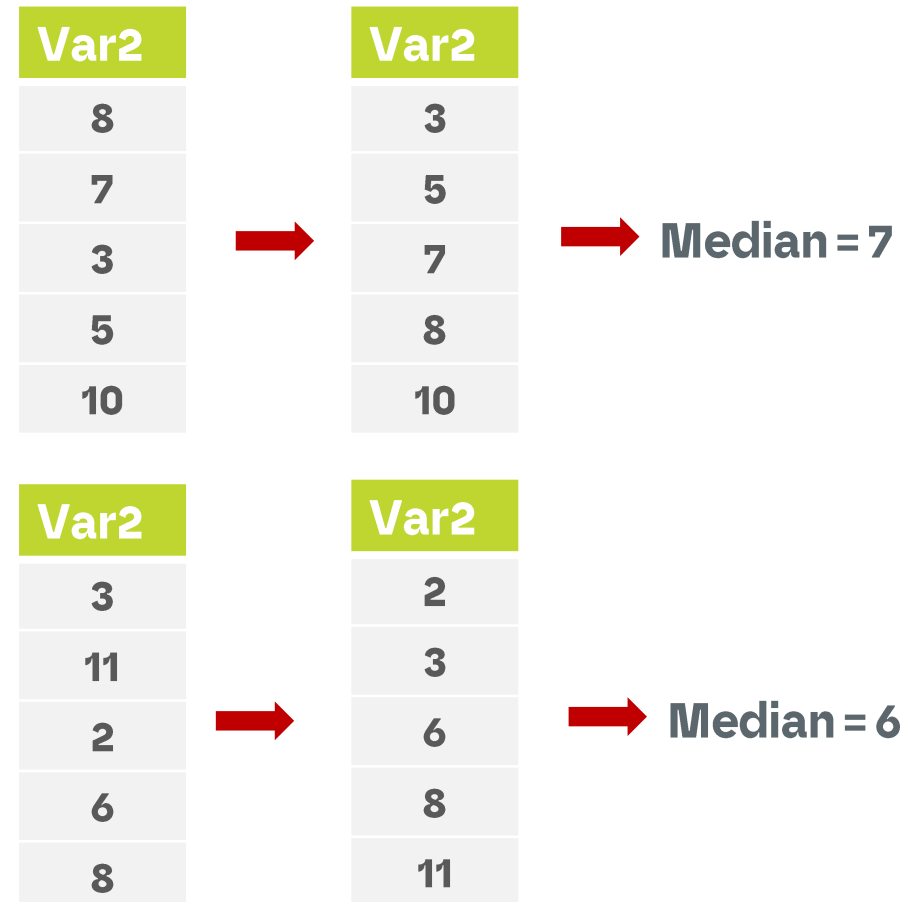
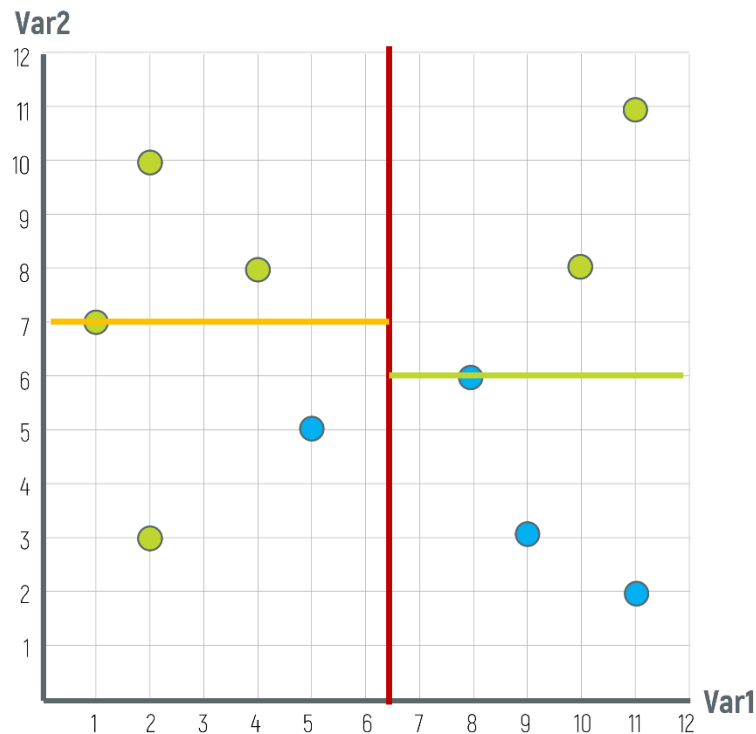
Var1	Var1
4	1
9	2
11	2
11	4
1	5
2	8
5	9
2	10
8	11
10	11

→ Median = 6.5

4.3. Instance Based Classifiers

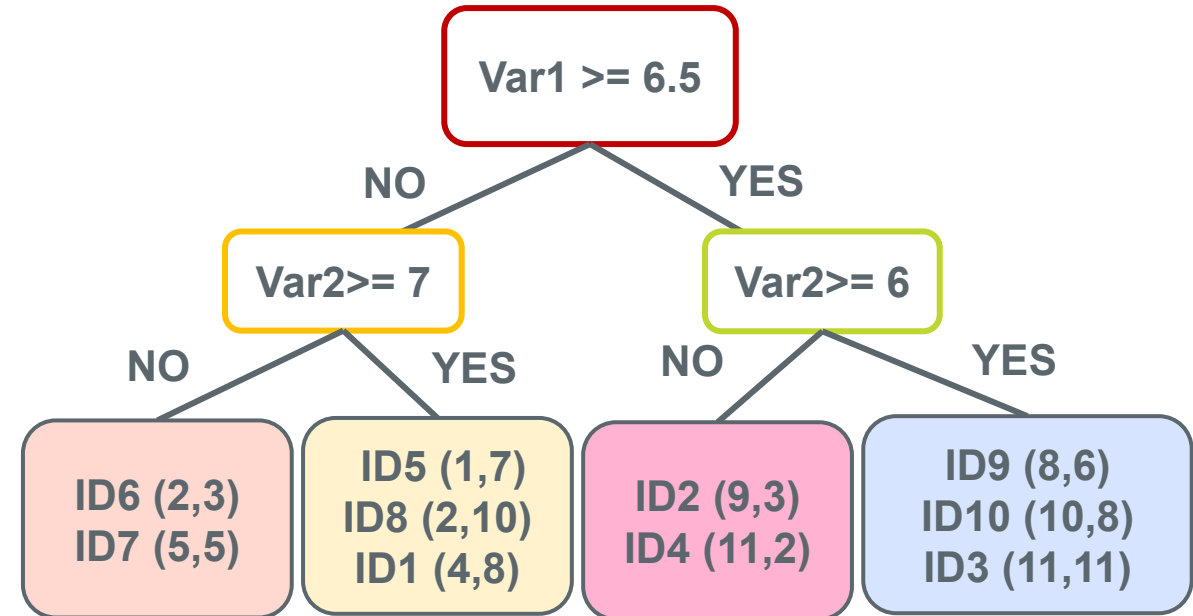
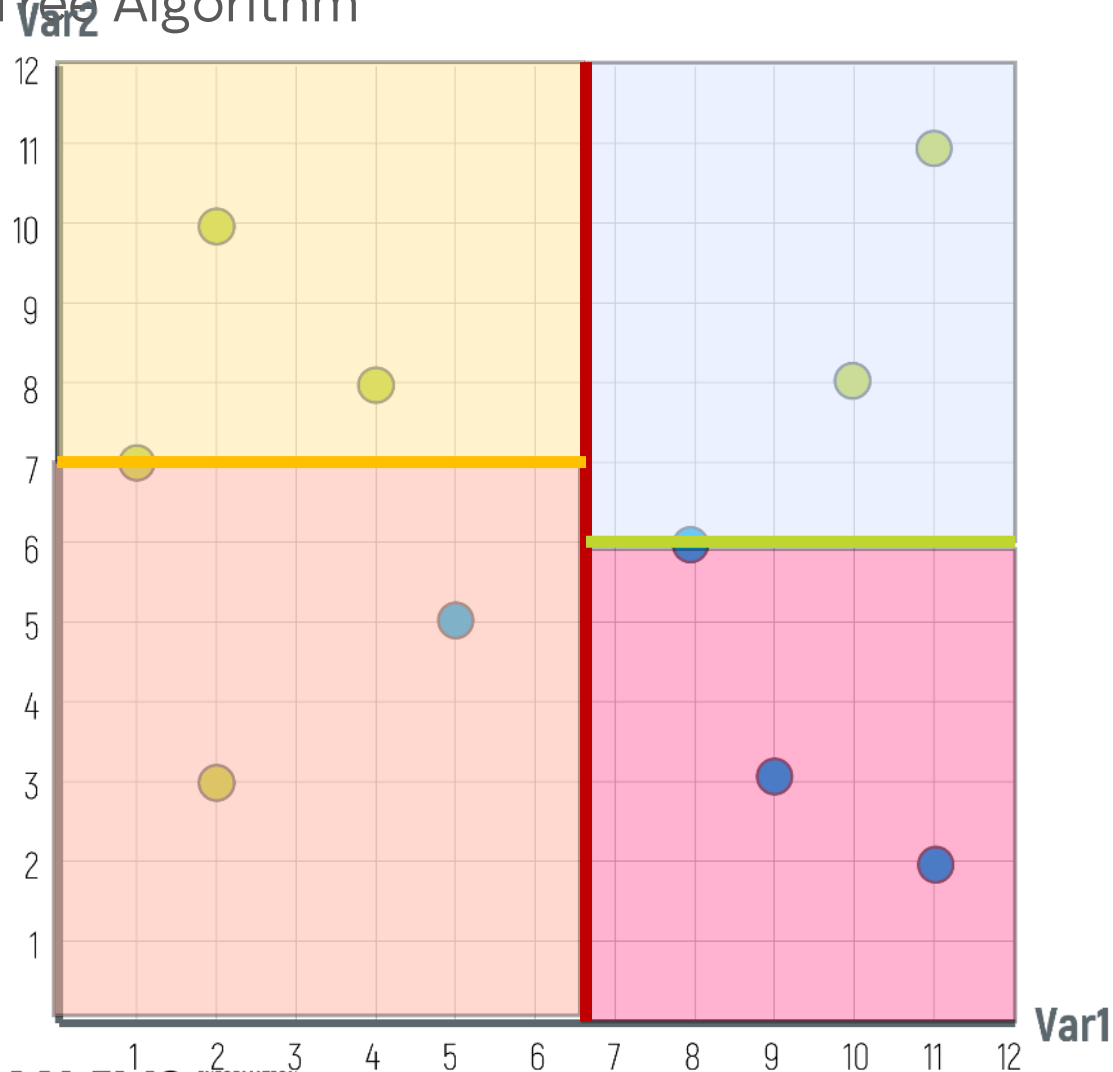
KDTree Algorithm

Step 2 – Go to the second feature, find the median and split accordingly. Do these splits (changing at each level the feature used) until the desired depth.



4.3. Instance Based Classifiers

KDTree Algorithm



4.3. Instance Based Classifiers

KDTree Algorithm - RECAP

- KDTrees partition the feature space so we can rule out whole partitions that are further away than our closest k neighbors
- The splits are axis-aligned, which not extend well to higher dimensions
- Usually the best solution for low-dimensional data

Obrigada!

Thank you!

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